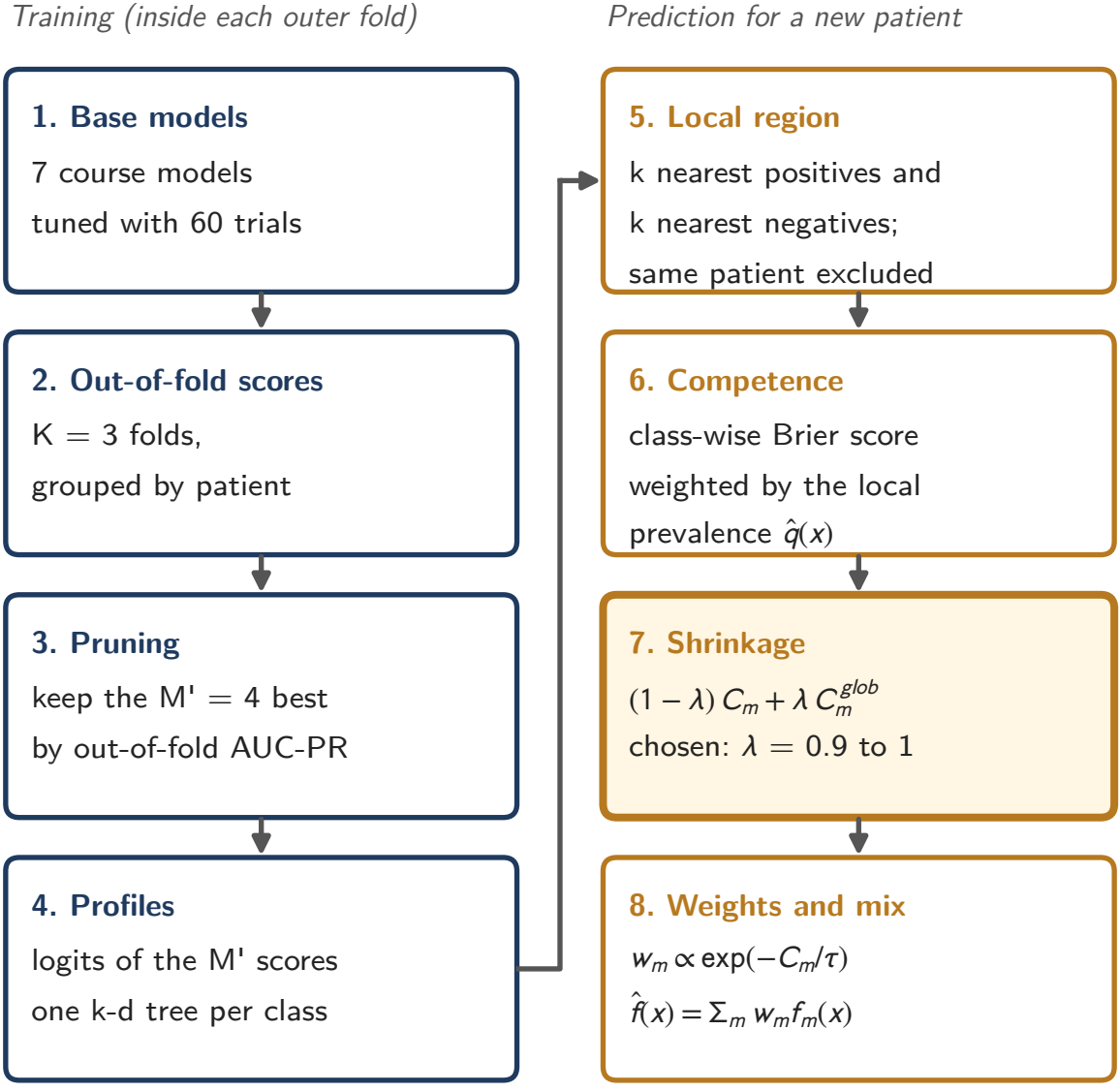
Training (inside each outer fold)
1. Base models
7 course models
tuned with 60 trials
2. Out-of-fold scores
K = 3 folds,
grouped by patient
3. Pruning
keep the M' = 4 best
by out-of-fold AUC-PR
4. Profiles
logits of the M' scores
one k-d tree per class
Prediction for a new patient
5. Local region
k nearest positives and
k nearest negatives;
same patient excluded
6. Competence
class-wise Brier score
weighted by the local
prevalence $\hat{q}(x)$
7. Shrinkage
$(1-\lambda)\, C_m + \lambda\, C_m^{glob}$
chosen: $\lambda = 0.9$ to 1
8. Weights and mix
$w_m \propto \exp(-C_m/\tau)$
$\hat{f}(x) = \Sigma_m\, w_m f_m(x)$